

Routing Concepts — Guided Notes

Unit 1.8.1 | CompTIA Network+ N10-009 Obj. 1.8 | N10-008 1.8

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

A router makes forwarding decisions by consulting its _____. Each entry contains the _____ network, the _____ IP or exit interface, and metadata including _____ and _____. If no match exists and there is no default route, the router _____.

Question 2

A routing table has entries for 10.0.0.0/8, 10.1.0.0/16, and 10.1.1.0/24. A packet destined for 10.1.1.75 will match _____ entries but be forwarded using the _____ route because of the _____ rule.

Question 3

Connected routes appear automatically when _____. They require no manual configuration and have an administrative distance of _____. A static route has an AD of _____, meaning it is trusted _____ than a connected route.

Question 4

Static routing is appropriate for _____ networks because it is _____. It fails to scale because _____. Dynamic routing solves this by _____.

Question 5

A default route uses the CIDR notation _____. It matches _____ destinations not covered by a more specific route. Home routers use a default route pointing to _____ so they don't need to know every route on the internet.

Question 6

Fill in the default administrative distance values:

Connected route → _____

Static route → _____

OSPF → _____

EIGRP → _____

RIP → _____

Question 7

OSPF uses _____ as its metric, derived from _____. RIP uses _____ as its metric with a maximum of _____ hops. RIP is considered inferior because _____.

Question 8 — Real World Application

REAL WORLD: A user reports they can reach some servers on the 10.2.x.x network but not others. You run 'show ip route' on the edge router and see a static route for 10.2.1.0/24 and an OSPF route for 10.2.0.0/16. A packet to 10.2.1.50 goes where, and why? A packet to 10.2.5.50 goes where, and why?

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/1-8-1-routing-concepts/1-8-1-  
routing-concepts-answer-key.pdf
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